ZIKRY ADJIE NUGRAHA

AI Engineer

Jakarta, Indonesia
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SUMMARY

AI Engineer specializing in AI, LLMs, Machine Learning, NLP, and Computer Vision, with 4 years of experience solving complex business problems for top national and international companies across the mining, e-commerce, and digital marketing industries. Highly proficient in Python, SQL, and major cloud platforms such as GCP, AWS, and Microsoft Azure for end-to-end development, spanning data engineering and modeling to backend architecture and frontend UIs. He is experienced in leading and deploying over 20+ impactful AI and data science projects, ensuring seamless go-live rollouts, MLOps integration, and actionable, real-life business solutions. Zikry is motivated to continuously broaden his expertise in driving cutting-edge AI innovations.

WORK EXPERIENCES

PT. Merdeka Copper Gold Tbk. (Leading Gold Mining Company in Indonesia)
Senior Full-stack AI Engineer – Specialized in building AI and ML Application for Geology and Mining-related project. **December 2025 – Present**

Key Responsibilities:
- Spearheaded end-to-end AI application development, focusing on backend architecture (APIs, databases, AI ML models), frontend UIs, and cloud deployment.
- Designed AI and ML algorithms to analyze geology and mining-related data to support data-driven decision-making and improve overall operational strategies.
- Developed frontend UIs (HTML, CSS, JavaScript) that translated complex data into actionable dashboards and applications for stakeholders and site engineers.
- Deployed production-ready applications and data automation across Google Cloud Platform (GCP) and internal company servers.
- Executed successful go-live rollouts to stakeholders, ensuring seamless adoption by employees across the parent company and its subsidiaries.

Go-live AI Data Science Projects:
- **AI Agentic Chatbot App:** Deployed enterprise RAG chatbot for employees, enabling instant AI-powered queries across 100+ regulatory docs via web & API.
- **Geotech ML Dashboard:** Real-time monitoring of 500+ sensor points with 30-day forecasts and AI summaries, cutting engineers' decision-making time significantly.
- **Limonite consumption App:** Replaced manual estimation with 80%+ accurate ML forecasts, cutting planning time for nickel site engineers with what-if simulations.
- **AI Lithology App:** Reduced drill core classification from 1 hour to 3 min/box via 2-stage automated CV model, eliminating inconsistent manual inspection at scale.

NucleusX BV (AI & Data Analytics consultant firms based in Netherlands)
AI Data Scientist – Specialized in AI LLM, NLP, and Data Analytics projects. **February 2024 – November 2025**

Key Responsibilities:
- Developed an end-to-end AI, ML, and NLP pipeline seamlessly integrated and deployed into Amazon Web Services (AWS) cloud infrastructure.
- Integrated Azure OpenAI models into client applications, optimizing LLMs and prompts for highly accurate classification, matching, and QA.
- Implemented MLOps standards (Docker, CI/CD) and project automation to streamline AI deployment workflows.
- Collaborated with international stakeholders and multidisciplinary teams, communicating complex data science insights to C-level leaders in business English.

Go-live AI Data Science Projects:
- **Multilingual RAG Chatbot:** Deployed 3-language (EN/DE/NL) AI chatbot for a European Fintech client, handling 100+ daily interactions via Azure OpenAI and Intercom.
- **AI FAQ Automation System:** Automated insight extraction from 300+ weekly FAQs using LLM-powered answering system, cutting manual support analysis time.
- **AI Market detection:** Eliminated manual categorization of 20,000+ company profiles, cutting classification time from 5 days to 1 hour via LLM pipelines.
- **AI Store Review:** Improved review classification 6x better with LLM that reduces total unclassified labels from 60% to 10% of 50,000+ multilingual (EN/DE/NL) reviews.
- **NLP Product Matching:** Raised product match rate from 50% to 85%+ via redesigned NLP algorithm, delivering more reliable competitive pricing comparison.
- **Product Standardization:** Established an AI NLP product matching standardization, reducing algorithm creation from 3 weeks to 5 days, used in 5+ projects.

Ilmu One Data (Digital Marketing & Data Analytics consultant firm based in Indonesia)
Data Scientist – Specialized in Data Analytics, Machine learning, and Computer vision projects. **March 2022 – January 2024**

Key Responsibilities:
- **(Acting senior)** Led the team projects, delivered and presented the data science business solutions to numerous national clients on the weekly basis.
- **(Acting senior)** Managed task distribution and project timeline among other team members based on their skills, knowledges and workload.
- Designed and monitored the ETL BigQuery, Python data transformation, and automation for numerous projects within Google Cloud Platform.
- Analyzed and gained business solutions for data-driven decision-making from Google Analytics and visualized it into Looker Studio dashboard.

Go-live AI Data Science Projects:
- **Computer Vision Ad Analysis:** Reduced ad creative analysis from 3 weeks to 1 day by automating visual feature extraction across 100+ monthly creatives.
- **ROI Forecasting ML:** Replaced manual reporting with an 85%+ accurate time series model, enabling 5-day advance spend planning across all channels.
- **Shrimp Pond Detection:** Mapped 100+ pond locations nationwide at 90%+ accuracy via YOLO satellite model, fully automating geographic surveying.
- **BigQuery Data Migration:** Reduced GA360-to-GA4 discrepancies from 50% to 10% for multiple e-commerce clients, ensuring reliable migration reporting.
- **Business Dashboard:** Automated daily GCP pipelines and Looker dashboards for 3 clients (e-commerce, FMCG, automotive), replacing manual reporting.

SKILLS AND KNOWLEDGE

- **Programming Skills:** Backend (Python, SQL, JavaScript), Web Frameworks (FastAPI, Flask, Streamlit), Frontend (HTML, CSS, Bootstrap), Cloud (GCP, AWS, Azure).
- **AI & LLM:** Azure OpenAI, Google Gemini, Vertex AI, Anthropic Claude, LangChain (RAG chains, multi-agent pipelines), Agentic AI, Prompt Engineering, AI Infrastructure.
- **Devops & Infrastructure**: Docker Compose, Traefik, GCP (Cloud Run, Artifact Registry), AWS (EC2, ECR, App Runner), Production Servers (Granian ASGI, Gunicorn).
- **Workflow Automation**: CI/CD, AWS Lambda, GCP Cloud Scheduler & Pub/Sub, AWS & GCP Secrets Manager.
- **Computer Vision:** Google Cloud Vision, GPT-4 Vision, OpenCV, YOLO, TensorFlow, Keras, Tesseract OCR, PIL/Pillow, scikit-image.
- **Natural Language Processing:** HuggingFace Transformers, Topic Modelling, NLP-based Similarity (Jaro-Winkler, Cosine Similarity, Fuzzy Matching), Regex.
- **Machine Learning:** Scikit-learn, XGBoost, Random Forest, ARIMA (time-series forecasting), BQML, Hyperparameter tuning, AWS Sagemaker, Azure ML Studio.
- **Data Storage:** Amazon S3 (boto3), Google Cloud Storage, SFTP.
- **Database:** Google BigQuery, Amazon RDS, PostgreSQL, SQL Server, SQLAlchemy, MySQL, Vector DB (Azure Cosmos DB, Qdrant).
- **Data Analytics:** Google Analytics (GA 360 / GA4), Google Ads Data hub, Google Ads Manager.
- **Data Visualization:** Matplotlib, Seaborn, Google Looker Studio, Tableau, PowerBI, Plotly Dash, Mapbox GL JS (interactive map).
- **Data Scraping & API Integration:** Selenium, BeautifulSoup, Marketing APIs (SerpAPI, SimilarWeb RapidAPI), Geospatial API (Mapbox API, Gmaps API).

LIST OF DEPLOYED PERSONAL PROJECTS

AI Enhanced News Sentiment Analysis Dashboard (Demo Application, Github)
- Automated Google News RSS scraper enriched by Gemini 2.5 AI for sentiment classification, NER, and topic tagging. Renders real-time dashboard with time-series charts and AI narrative. Deployed on GCP Cloud Run.

AI-Driven Diabetes Prediction and Personalized Nutritional Recommendation System (Demo Application, Github)
- Scikit-learn diabetes classifier with KNN-based 3-meal nutritional recommendations and multimodal food-label OCR, each enhanced with Gemini GenAI for personalized nutritional insights. Deployed on GCP Cloud Run.

AI-Skill Gap Analysis (Github)
- 4-agent LangGraph pipeline parsing CVs against live job market data via Google Search grounding. Outputs structured markdown upskilling report covering skill gaps, strengths, and recommendations for any target role.

AI-Driven Export (Github)
- Multimodal OCR pipeline extracting HS codes from product images via LLM, paired with 5-year UN Comtrade trade analytics identifying top 3 Asian import markets. Deployed on AWS ECS + ALB with PostgreSQL token quota system.

AI-Powered Alarm Diagnostic Assistant: Integrating BLIV and Vertex AI for Smart Industrial Maintenance with RAG (Github)
- Integrates BLIV Dashboard real-time monitoring with Vertex AI RAG, indexing industrial maintenance PDFs into BigQuery vector store, retrieving top-3 relevant chunks with Gemini grounded answers to diagnose machine errors with matching guidebook page visuals and optimize maintenance resource utilization.

LIST OF ARTICLES AND RESEARCH PUBLICATIONS

Evaluation of the Multimodal Gemini Model in Analyzing Nutritional Value Information of Products for Diabetes Prevention (Journal link, Seminar proof)
- Evaluates the Gemini 1.5 Flash multimodal LLM model in analyzing sugar content from photos of nutritional information on 10 beverage products, achieving 90% accuracy and an F1 score of 94%.

Simple Computer Vision Image Creative Analysis using Google Vision API (Github, Towards Data Science Article)
- Tutorial demonstration on using Google Vision AI to extract meaningful information from an image and identify key parameters such as object, text, color.

Tree-Based Algorithms Approach on Predicting Customer Satisfaction (Github, Towards Data Science Article)
- Tutorial on designing a machine learning hyperparameter model and evaluating it to predict customer satisfaction from an e-commerce dataset.

EDA and Visualization of Indonesia Earthquake Over the last 20 years (Github, Towards Data Science Article)
- Earthquake data analysis, visualization, and insights creation focused on the region of Indonesia over the past 20 years.

EDUCATION

Sultan Ageng Tirtayasa University — **September 2015 – July 2019**
- **Major:** Bachelor of Metallurgical Engineering (GPA 3.28 of 4.00)
- **Data Science related study:** Calculus, Statistics, Physics, Numerical Analysis & Modeling, Computer & Programming, Metallurgical Process Simulation

TRAINING AND CERTIFICATIONS

IELTS Academic Certification — **July 2025**
- **Score:** Overall 7.0, Listening 8.0. Reading 7.5, Writing 6.5, Speaking 6.0

Google Analytics Full Course for Certification — **March 2022 – May 2022**
- **Major:** Google Analytics Certification course
- **Data Science related study:** Advanced Google Analytics, Google Looker Studio, Google Analytics 4 and 360, Google BigQuery

Dibimbing IT Bootcamp — **August 2021 – December 2021**
- **Major:** Full stack Data Science Bootcamp
- **Data Science related study:** Python, SQL, Data Science theory, Exploratory Data Analysis, Data visualization, Machine Learning, Project preparation

ACHIEVEMENTS AND AWARDS

Top 8 finalist of Spark Telkomsel National Datathon (News 1, News 2, LinkedIn) — **February 2025**
- Surpassed 65+ teams nationwide, recognized as the only team who delivers the only AI innovation in manufacturing.

Top 10 finalist of AI category in BPJS Kesehatan National Healthkathon (News, LinkedIn). — **November 2024**
- Chosen as a finalist from 350+ teams across national level, recognized for developing an AI innovation for the Diabetes patient.

Excellence Staff in Domain Knowledge Curiosity IlmuOne Data — **March 2023**
- Acknowledged as the top performer among 30+ coworkers for outstanding curiosity in domain knowledge exploration.

2nd best Data Science Final Project — **December 2021**
- Achieved 2nd best project among 40+ students with project featured on Towards Data Science platform.

Grantee of the POSCO Asia International Fellowship Scholarship — **October 2018**
- Earned the scholarship for self-development and thesis after a highly competitive selection process involving over 100+ applicants in Indonesia.